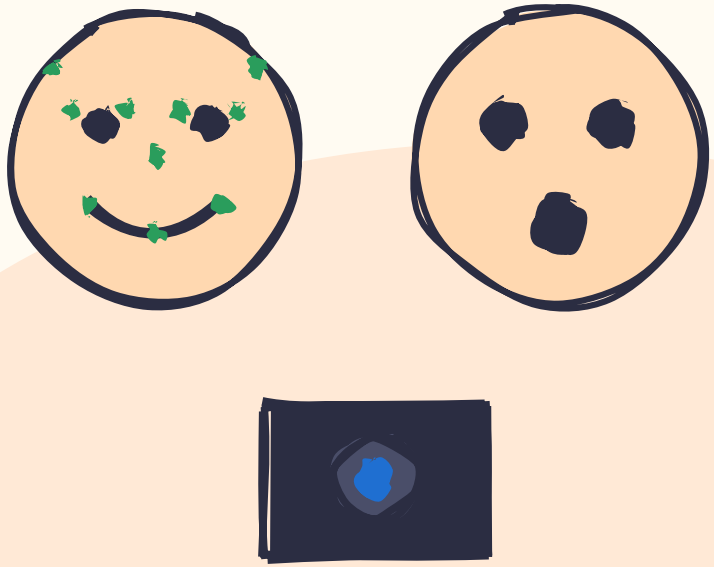
Junior Makers · Class 31

AI Emotion Mirror

Teach a computer to read your face

Face landmarks | Training a model | Reacting sprite | Fairness & privacy



Did you know?
Your phone can unlock just by seeing your face. It measures the shape of your face, not just a photo.

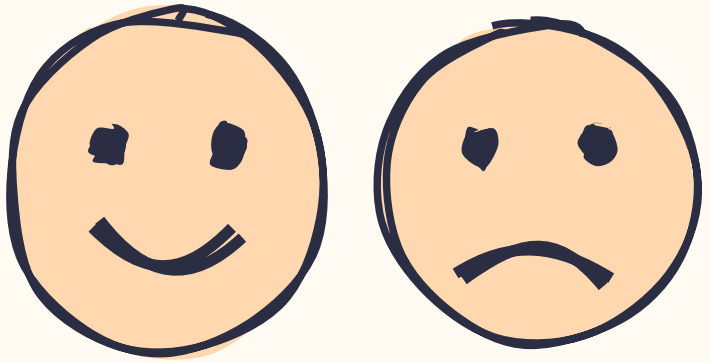
LEARN

Let's think...

Can a computer tell if you're happy?

A smiling camera

Some cameras only take the photo when everyone is smiling. Some apps put silly filters exactly on your nose. The computer has no feelings, so how does it 'see' a smile?



It feels happy too?

It guesses?

It learned from examples!

Today: you **train an AI model** on your own expressions and build a mirror sprite that reacts to your face.

Learning objectives

By the end of this class I can...

1	**How AI sees:** explain how a computer finds **faces and expressions** from points and examples	**Proof:** explain 'landmarks' with the face map
2	**Train a model:** train an **image classifier** with labelled examples in PictoBlox	**Proof:** my model tells happy from surprised
3	**Mirror app:** make a sprite **react** to my expression	**Proof:** my sprite changes when I smile
4	**Fair & private:** test the model on **different people** and discuss privacy and fairness	**Proof:** our fairness test table and one improvement

Objective 1 · How AI sees · video

Video time!



How Does Facial Recognition Work?

Hashem Al-Ghaili · about 4 minutes · youtu.be/agGEDdj05U0

Watch for...

1. What **points** on a face does the computer measure?
2. How does it compare a new face with faces it knows?
3. Where is face recognition used in real life?

Scan to watch

After the video: Is it OK for a camera to recognise you without asking? Discuss with your partner.

Objective 1 · How AI sees

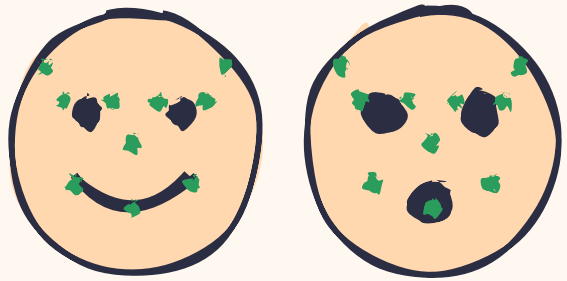
Faces are points and patterns

The idea

First the computer **finds the face** in the picture. Then it marks **landmarks**: corners of the eyes, tip of the nose, corners of the mouth.

A smile pulls the mouth corners **up and apart**. Surprise opens the mouth and eyes **wide**. The computer learns these **patterns** from lots of examples.

Example: drawing emoji



With just two dots and a curve you can draw happy 🙂 or sad 🙁. Our brains read faces from a few key points. The computer does the same, with numbers.

Objective 2 · Train a model

Training: examples + labels

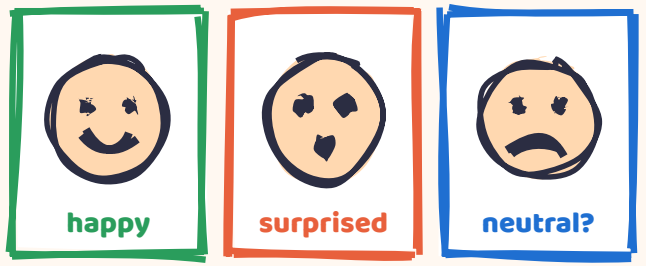
The idea

An **image classifier** learns from **classes** (groups) of example pictures:

- Class 'happy': 30 pictures of smiling faces
- Class 'surprised': 30 pictures of surprised faces
- Class 'neutral': 30 pictures of normal faces

It finds what makes each class different. Then it gives a **confidence** for new pictures: 'happy 92%'.

Example: a baby learning animals



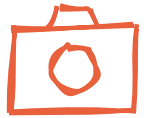
A toddler sees many dogs and hears 'dog', many cats and hears 'cat'. Soon they can name a dog they've never seen. Nobody told them the rules: they learned from examples.

Real world

Face AI around us



Face unlock
Phones recognise their owner's face



Smile shutter
Cameras snap when everyone smiles



Filters
Apps put ears and glasses exactly on your face



Helping people
Apps help people with autism practise reading expressions

Which of these uses would you be happy with? Which would worry you?

Face vs emotion
Recognising WHO you are and guessing HOW you feel are different jobs, and emotion guessing is much less reliable.

Objective 4 · Fair & private

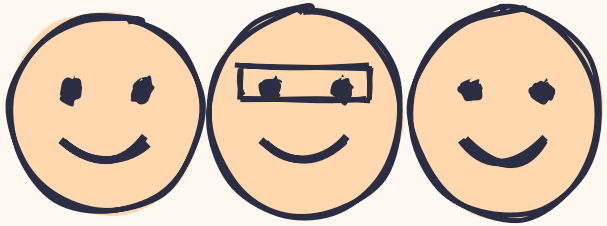
Does it work for everyone?

The idea

Our model learned from **a few faces in our class**. Test it on people it has **never seen**: different skin tones, glasses, a cap, in dim light, from far away.

If it fails for some people, that's **bias**: the training examples didn't include enough variety. The fix is **better data**, not blaming the person!

Example: learning only one accent



glasses? light? distance?

If you only ever heard English from one teacher, you might struggle to understand someone from another country. Not because they speak wrong, but because your 'training' was too narrow.

Objective 4 · Fair & private

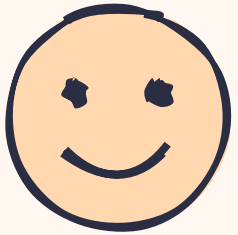
Expressions are not feelings

The idea

The AI only sees the **outside** of your face. You can smile while nervous, or look 'neutral' while very happy inside. People from different families and cultures also show feelings differently.

So an emotion AI is making a **guess about a face**, not reading your mind.

Example: an actor



smiling...

but nervous?

Actors can look terrified on screen while feeling perfectly calm. If a camera can be fooled by acting, it can be wrong about real people too.

Part 2 · Make it

Now let's put it to work!

You know the ideas. Time to play, build and test. Use these slides while you work:

1. Human classifier
2. Collect your examples
3. Train and test the model
4. The emotion mirror
5. Fairness test
6. Bug hunt
7. AI challenges

Class game

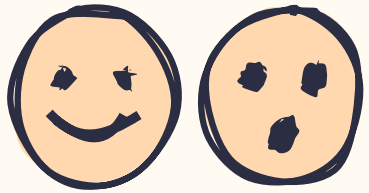
Human classifier

How to play

1. One student is the **AI**. They only learn from examples: no explaining!
2. The class shows 3 photos (or faces) labelled **HAPPY** and 3 labelled **SURPRISED**.
3. Now show 5 NEW faces. The 'AI' must guess each label.
4. Then try a tricky one: a face that is half smiling. What does the AI say?

Then discuss

How did the 'AI' decide? What would help it get better?



Big idea

AI learns patterns from labelled examples. More, varied examples = better guesses.

Objective 2 · Build step 1 of 3

Collect your examples

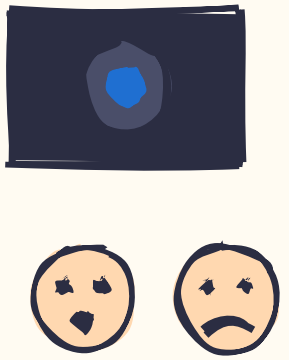
a Open **PictoBlox** → Block Coding → **Machine Learning Environment** → Image Classifier.

b Make 3 classes: **happy**, **surprised**, **neutral**.

c Use the webcam: record about **30 pictures per class**.

d Move a little: turn your head slightly, move closer and further.

e Ask a teammate to add pictures too, so it learns more than one face.



Check: Each class has around 30 varied pictures.

Only photograph people who say yes. Delete the pictures at the end if they ask.

Objective 2 · Build step 2 of 3

Train and test the model

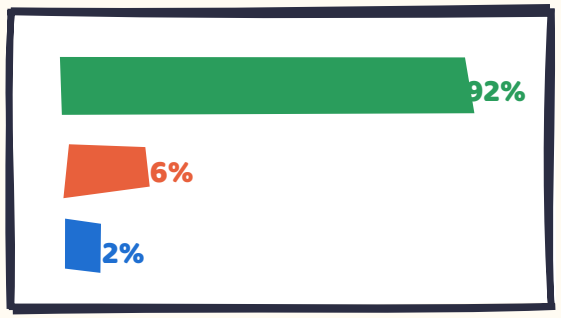
a Click **Train Model**. Wait for the progress bar.

b Look at the **accuracy** graph: it should rise as it learns.

c Click **Test**: make each face. Watch the confidence bars.

d Does it get confused? Add more pictures to that class and train again.

e When it's good, click **Export Model** to use it in blocks.



Check: Making a happy face shows 'happy' with high confidence.

Objective 3 · Mirror app · Build step 3 of 3

The emotion mirror



Confidence

Only react if confidence > 70%. Otherwise the sprite flickers between moods!

Your twist

Add a 'sad' class and make the sprite cheer you up.

Objective 4 · Fair & private · test it

Fairness test

Test	Right? (✓/✗)	Confidence
Teammate (in training)		
Friend NOT in training		
Wearing glasses / cap		
Dim light		
2 metres away		

Improve it

Add pictures for the cases that failed, retrain, and test again. Did it get fairer?

Test like an engineer

Bug hunt

Problem	Likely cause	Did it happen?
Always says 'happy'	one class has many more pictures	
Flickers between moods	no confidence threshold	
Works for me, not my friend	training only used my face	
Camera not found	webcam permission blocked	
Very slow	analyse less often (add wait 0.2 s)	

Debug checklist

Keep the classes balanced: about the same number of pictures in each.

Good lighting on your face (not behind you) helps the camera a lot.

Level up!

AI challenges

Choose your challenge



Gesture control: train thumbs up / thumbs down / wave to control a game.



Mood music: play a different song for each expression.



Fair model: collect pictures from 6 classmates and beat your fairness score.



Face game: the sprite shows an emoji; you must copy it within 3 seconds!

Think like an engineer

Why did adding pictures of more people make the model better?

Boss level

Use the model to control your Arduino: LED green for happy, red for sad!

Think about it

Your face is your data

Questions to ask

- Who can see the pictures the app takes?
- Are they stored, or deleted?
- Did everyone in the picture agree?

Debate

A school wants cameras that guess if students are 'paying attention' by reading faces. Good idea? What could go wrong?

Our rule

PictoBlox trains on YOUR computer. At the end, delete the training pictures if anyone asks.

MAKE

Rules vs learning

Two ways to make a computer smart

Rules (normal code)

IF mouth corners go up THEN happy.
We write every rule ourselves. Hard for faces: there are too many ways to smile!

Machine learning

Show lots of labelled examples. The computer finds the rules itself. Great for messy things like faces, voices and handwriting.

Objective check

Can I really do it? Prove it.

01	What are face landmarks? Point to 3 on the face map.	○ ○ ○
02	How many classes and pictures did you use? Show the confidence when you smile.	○ ○ ○
03	Show your mirror sprite reacting to 2 different expressions.	○ ○ ○
04	Read your fairness test. Who did it fail for, and how did you improve it?	○ ○ ○

Rate yourself

Colour one circle each. Red: not yet. Amber: with help. Green: I can explain it to a friend.

Teacher: Mirror show: each team demos its sprite and shares one fairness finding.

Show what you know

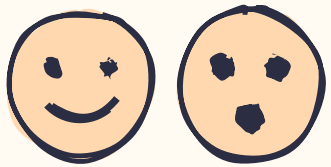
What did we learn?

- [x] AI finds faces and landmarks, then learns patterns
- [x] Training = labelled examples in balanced classes
- [x] Confidence tells how sure the model is
- [x] Narrow training data causes bias: test on everyone
- [x] Faces are personal data; expressions aren't feelings

Exit ticket: your model works for you but not your friend. Why, and how do you fix it?

At home

Watch an advert or film with the sound off. Can you guess feelings from faces? Were you always right?



Next class: Cyber Safety Escape Room!